# Danish Residential Housing Prices

## Data Visualization

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## Group 16: Team Members

Platon Dimitriadis, pldim24@student.sdu.dk
Md Al Imran Khan, mdkha24@student.sdu.dk
Jacob Bregndahl Larsen, jacla20@student.sdu.dk

## Contribution

The following table shows the student's percentage of contribution.

| Student | Percent of contribution (PoC) |
|---|---|
| Platon Dimitriadis | 100 |
| Md Al Imran Khan | 100 |
| Jacob Bregndahl Larsen | 100 |

## Application

The Shiny application has been published and can be accessed here:
https://j4rb.shinyapps.io/Danish-Residential-Housing-Prices/

## Abstract

This project explores the Danish residential housing market from 1992 to 2024 using interactive data visualizations. Based on a 100,000-row sample dataset, we designed and implemented an R Shiny dashboard to examine trends in housing prices, house types, regional differences, and buyer behaviour. A wide range of visualization techniques, including histograms, box plots, bar charts, heatmaps, time-series plots, maps, and animations, were used to answer research questions.

The visualizations reveal an upward trend in housing prices, a consistent dominance of villas as the most purchased housing type, significant regional price disparities favouring urban and coastal areas, and notable relationships between house size, number of rooms, and pricing. Price negotiation patterns indicate that final sale prices are on average lower than listing prices, though with substantial variation across time and regions.

# 1. Background and Motivation

Our team started researching for the acquisition of a dataset that can be used for the development of our project. We have evaluated many good candidates that could be used as our core dataset, but we decided to go with the Danish Residential Housing Prices between 1992 and 2024. Housing prices change very often between decades, sometimes increasing and decreasing accordingly. The reasons for these fluctuations, as they do in many countries, vary a lot, including geopolitical reasons, economic cycles, and changes in living standards.

Denmark is an especially interesting country to research. It has high living standards, a good and contained economy, and an organised society. Denmark is divided into small regions, for which the dataset also contained information. Therefore, it was a suitable candidate to research house prices and examine the stability of the prices. We believe that we can deduce a story behind Danish housing price fluctuations by visualising and analysing the data from many distinct types of visualizations (both geographic visualizations and pure graphical ones).

We believe that through the visualization of the entries from this dataset, we can gain insights that can help the readers gain insight into how and why these fluctuations occur. The dataset contains an enormous number of entries and many different variables, which can create great opportunities for visualizations.

# 2. Project Objectives

When we were searching for a dataset, we were looking for meaningful data that has the potential to answer some basic questions about today's market, especially the housing market. Because, in recent years, buying a house has become more of a headache as to what kind of house one should purchase, if the location is ideal, or even if one should purchase immediately or wait more years to observe the market, and if the price is low enough, then they could purchase it. All these questions raise our awareness when we enter the housing market, not only as an ordinary family but also as a potential real estate agent. Bearing all these objectives in mind, we found this dataset.

Denmark is one of the countries in the world that is not exceptionally large in terms of habitable places. In fact, it is ranked 130$^{th}$ in the world. [1] Still, there are multiple big cities from which we could choose when selecting our next destination to live in. Some cities are too expensive, while others have a range of house types to choose from.

All these lead us to formulate the following questions that we will answer through this visualization project.

- ❖ What are the most popular types of houses on sales in Denmark?
- ❖ What is the most purchased type of house in Denmark in the past decades?
- ❖ Does the house purchasing uprising or downsizing through the years in Denmark?
- ❖ Does the number of rooms affect the sales of houses in Denmark, and if so, then how?
- ❖ What was the house price index through the years?
- ❖ How do square-meter prices vary by area for Danish residential housing?
- ❖ Does the negotiation play a crucial part when purchasing a house in Denmark?

# 3. Data

This section describes the data used to create the visualizations and the preprocessing steps we took to prepare the data.

## Source

The dataset used in this project comes from Kaggle and is called *"Danish Residential Housing Prices 1992–2024"* by Martin Frederiksen. [2]

For this project, we used the 100.000 sample version of the dataset provided in the repository. The sample contains enough data to be representative while being a manageable size.

## Description

The dataset contains transactional records of residential property sales in Denmark between 1992 and 2024. Each row represents a single property sale.

The variables included in the dataset are:

| **Variable** | **Description** | **Data Type** |
|---|---|---|
| date | The date on which the property transaction took place. | numerical, discrite |
| quater | The calendar year quarter corresponding to the sale date. | categorical, ordinal |
| house_id | A unique identifier for the property. | categorical, nominal |
| house_type | The category of residence, such as Villa, Farm, Summerhouse, Apartment, or Townhouse. | categorical, nominal |
| sales_type | The type of sale, such as regular sale, family sale, other sale, or auction. | categorical, nominal |
| year_build | The year the property was originally constructed. | numarical, discrite |
| purchase_price | The final purchase price of the property in DKK. | numarical, continuous |
| change_between_offer_and_perchage | The percentage difference between the listing offer price and the final purchase price. | numarical, continuous |
| no_rooms | The total number of rooms in the residence. | numarical, discrite / categorical, ordinal |
| sqm | The size of the property measured in square meters. | numarical, discrite |
| sqm_price | The price per square meter, calculated as purchase_price divided by sqm. | numarical, continuous |
| address | The street address of the property. | categorical, nominal |
| zip_code | The postal code where the property is located. | categorical, nominal |
| city | The city or town where the property is located. | categorical, nominal |
| area | The broader area classification of the location within Denmark. | categorical, nominal |
| region | The geographical region of Denmark in which the property is located. | categorical, nominal |

| | | |
|---|---|---|
| nom_interest_rate | The nominal Danish interest rate reported on a quarterly basis. | numarical, continuous |
| dk_ann_infl_rate | The Danish annual inflation rate reported for the quarter. | numarical, continuous |
| yield_on_mortgage_credit_bonds | The 30-year mortgage credit bond yield representing general financing conditions. | numarical, continuous |

## Data Processing

Only minimal preprocessing was required before visualizing the data. All datasets were loaded in global.R to avoid repeated reading during the Shiny session. The spatial datasets containing Danish postal codes and region boundaries were imported from GeoJSON files, and their key attributes were converted to character type to ensure correct matching with the housing data.

The housing dataset was loaded from CSV, and two transformations were applied. The date column was converted to a proper Date object, and a new year variable was extracted from the date.

# 4. Visualization/Dashboard

## Technology

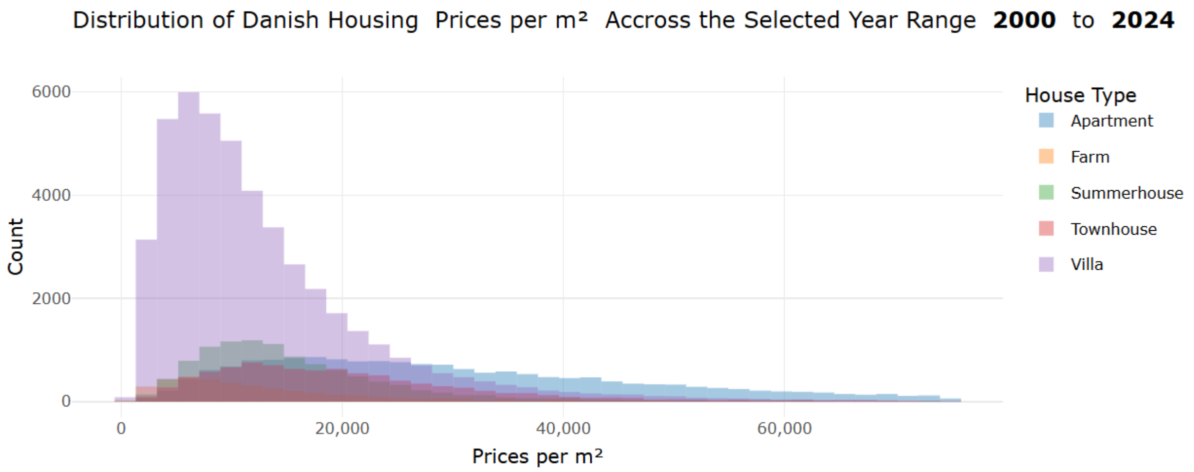
We are using Shiny for our visual encodings, as Shiny is easy to use and it uses an R package, which makes it easy to build interactive web apps. [3] As for designing our dashboard, we have chosen “Vanilla Shiny” as it is simple and fast to build, stable, and well supported with flexible HTML and CSS. It also has very few dependencies and is, as a result, easy to fit with Shiny modules. Though it has some disadvantages also. For example, it uses Bootstrap 3 with limited built-in theme support. But nevertheless, we prefer this because of its simplicity and as a basic iteration of development.

## Design

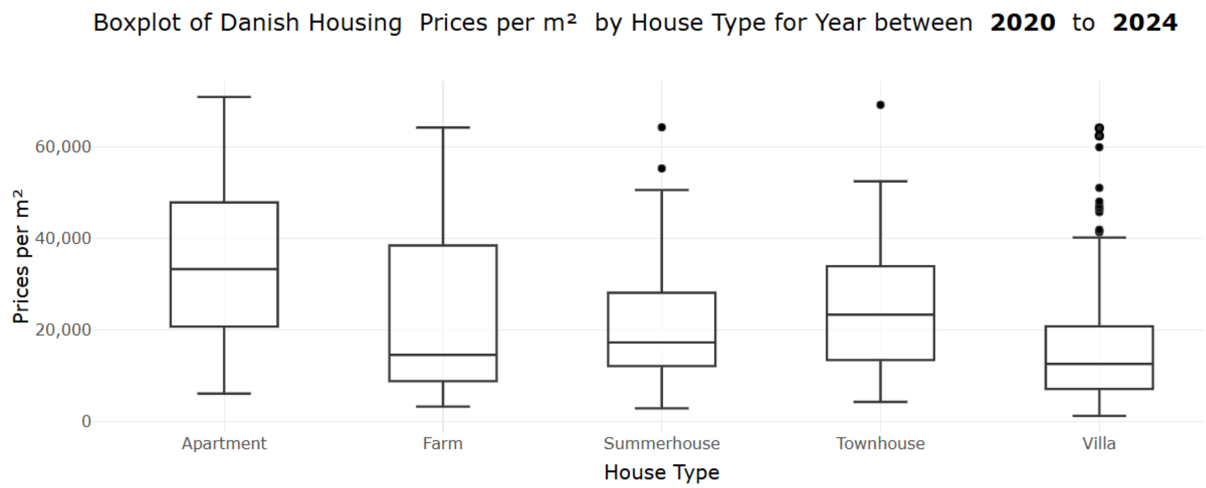
In this visualization, we mainly focus on different graphs like histograms, box plots, bar charts, scatter plots, line graphs, heatmaps, animations, and maps to demonstrate the overall picture of the Danish housing market’s purchase history and how the market’s various aspects actually evolved from 1992 to 2024. These visualizations focus on the number of purchases, purchase price, per square meter price, purchase year, build year, and area of the purchases with regional information in Denmark. We also put emphasis on the type of houses, the number of rooms, and negotiations while purchasing these houses.

## Purchase Trend

In this tab, we examine the trend of housing prices, purchasing criteria, and, in this way, we answer some important questions. This tab shows 5 types of graphs. The first one is a histogram that explains the distribution of types of housing purchased from 2000 to 2024. We have a filter section where one can choose different combinations to observe the graphs as we examine several aspects of the housing market. There is a feature to either show the visualization or not by pressing the “Render/Hide Histogram” button. The plot value can be changed from “Price per $m^2$” to “Purchase Price.”

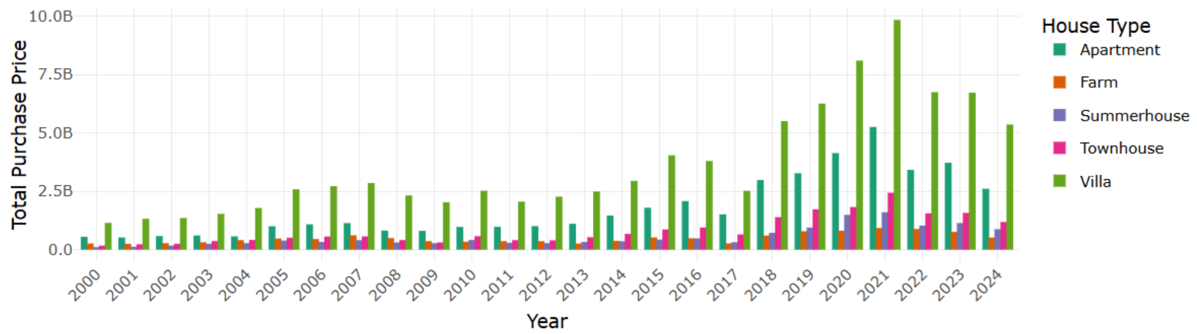



The next graph is a box plot of the different house types in the Danish market. By default, this graph is not rendered. To view this graph, we must press the “Render Box Plot” button. Here we can also see the graph by using the “Price per $m^2$” or “Purchase Price.” This graph analyzes the different prices and how they are distributed; for example, the price is not evenly distributed for farmhouses, whereas apartments have an even distribution.



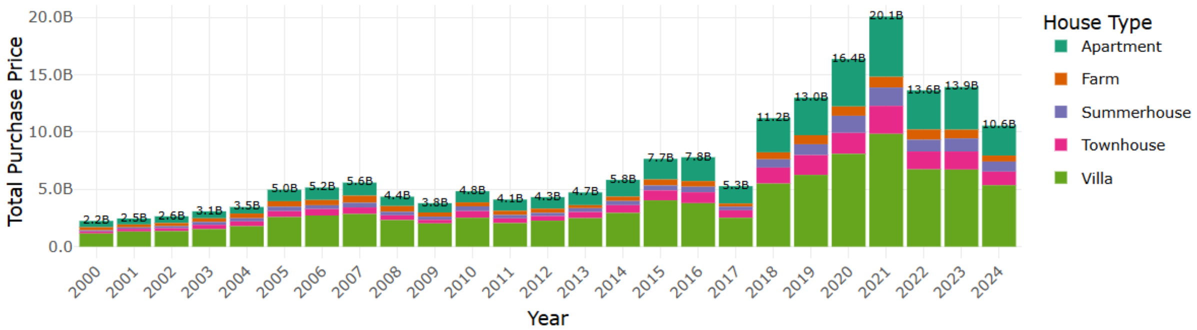

In this bar chart, we analyze the distinct types of houses that are purchased in different years side by side. From the filter, we can filter different selections to observe different combustion. Here we can plot the chart as either “Purchase Price” or “No. of sales.” Also, we need to click the “Render Bar Plot” button when first visiting the dashboard.

Position Dodge Barplot of Danish Housing Purchase Price for Year between **2000** to **2024** for Different House Type



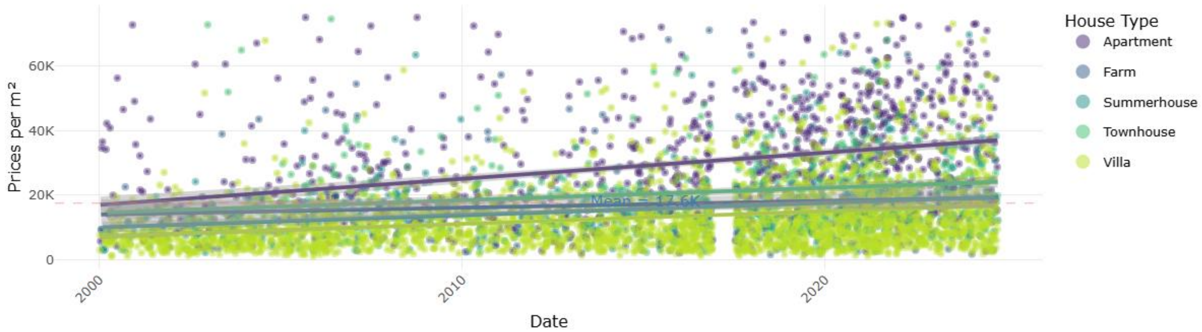

We can plot this bar chart as a stacked bar chart by changing the selected list from “Plot bar chart.” Here are all the same properties as described above, but in a stacked format.

Stacked Barplot of Danish Housing Purchase Price for Year between **2000** to **2024** for Different House Type




These graphs provide us crucial information about the trend of purchases over the year and how it is going now. Also, we could have some general idea about the preferred house types that were bought over the years. This is the last graph in this tab; it is a scatter plot, and to render it we need to press “Render Scart Plot,” because by default it is invisible to reduce the page loading time when someone visits the tab for the first time.

Scatterplot of Danish Housing Prices per m² of Year between **2000** to **2024** for House Type (Mean: 17.6K)

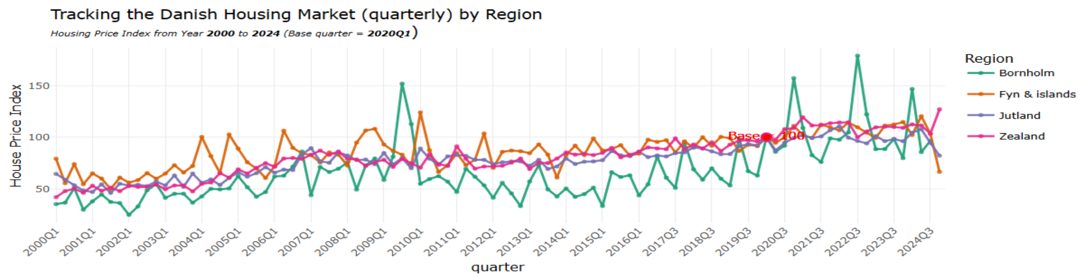


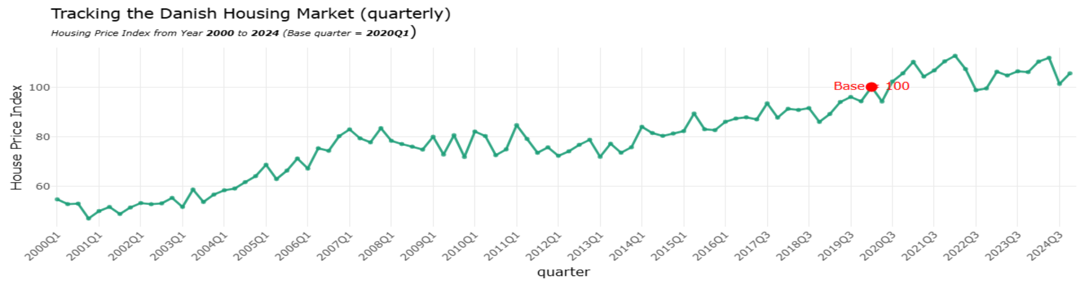

This graph is a general idea of the distribution of all the housing purchases throughout the year. As shown in the graph, the villa is the dominating figure, but it has a lower purchase price than the others in this category. This might be one of the reasons why it is one of the most popular types of housing when it comes to purchasing one.

All the graphs in the Purchase Trend tab have rich filtering criteria and flexible options for different plotting options. It also can observe these graphs using colours that are comfortable with colour blind visitors.

## Price Index

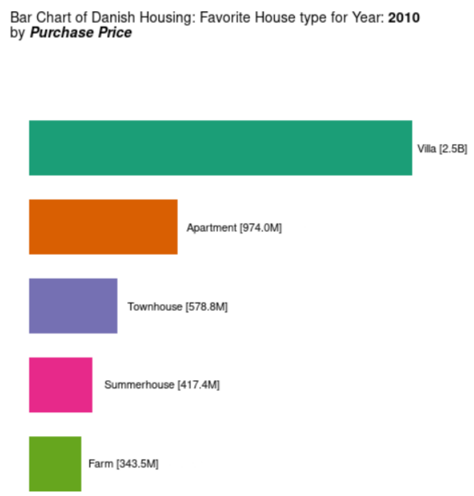
In this tab, we answer one of the most important questions, namely, how the price index was over the past 33 years. Here we present two types of graphs, the first one is the overall price index in Denmark, and the second one is the price index for the different regions in Denmark.




These two graphs demonstrate that, based on 2020Q1, the price increased. Here, we can observe the graph by quarter or by year. We can also select the base year or quarter by which we could analyze the price index. Another important feature of this graph is that we can plot it as a bar chart by selecting “Bar Chart” from the Show graph type filter criteria.

## Animation




In this tab, we are generating an animation from 1992 to 2024 of five house types. The main purpose of this animation is to answer one of the questions of which is the most popular house type was purchasing for every year. And according to the graph, it is obvious that Villa has dominated most of the years, while others have alternated their positions in some years.

## Map

An interactive map visualization has been developed using the Leaflet library. Its primary purpose is to allow users to explore the spatial distribution of housing market metrics across Denmark.

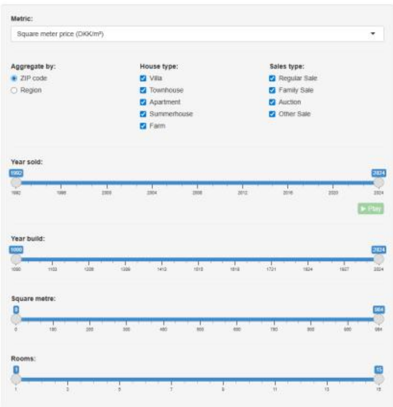
The map visualization was developed because location is a central attribute of housing data. Users can immediately identify geographic patterns and outliers that would be difficult to detect in non-geographic visualizations. We chose to make the map interactive to empower the users to explore their own questions about the Danish housing market.

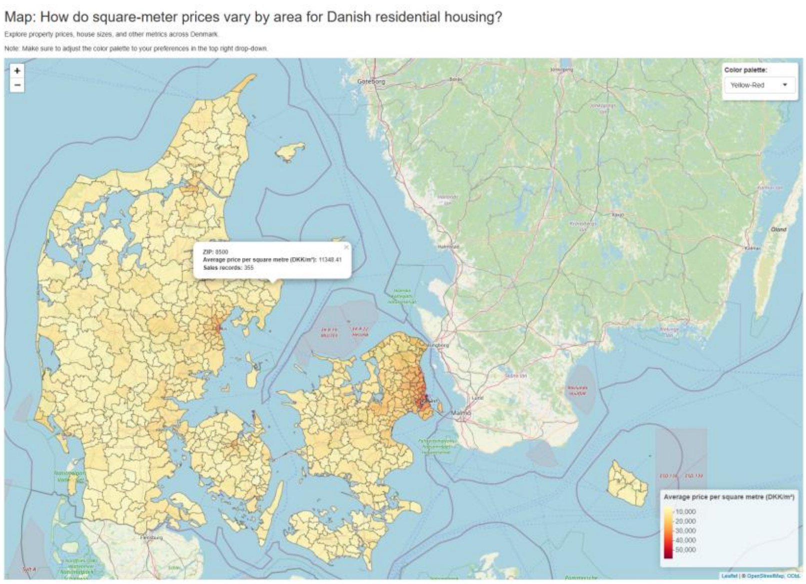
The map visualization works by layering .geojson files on top of the Leaflet map. The .geojson files contain polygons that outline geographical areas. Each polygon is coloured according to the mean of the selected metric, calculated for every house sale record within said geographic area. The .geojson files were acquired from the GitHub repository Neogeografen/dagi.

When a polygon is clicked, a pop-up containing contextual details, such as area name, the average metric value, and the number of sales records, is shown. This allows the user to see details for any given area while maintaining a clean visual overview.

The sidebar allows users to filter and customize the visualization dynamically. The map updates in real-time when any filter is adjusted. Controls include:

- Metric: Determines which housing attribute to visualize (price per square meter, total purchase price, house size, number of rooms, or year built)
- Aggregation level: Toggles between ZIP-level or region-level aggregation.
- Filters: Enables the user to subset the data by house type, sales type, year built, year sold, size, and number of rooms.
- Animated time slider: The year sold filter includes an animation to show the change in metric over time.
- Color palette selector: On the top right of the map, a drop-down with various color palettes can be accessed for accessibility.



There are several ways the map visualization could be improved. One enhancement is to geocode the individual property addresses in the dataset. This would allow us to expand the dataset with columns for municipality and longitude and latitude coordinates for every record.

Using these new columns, the map could be expanded to support municipality-level aggregation in addition to the current ZIP-code and region layers. This would give users an intermediate spatial unit that is often more meaningful for policy, planning, and public discussions of the housing market.

With latitude and longitude coordinates, the application could also introduce a bubble map instead of relying solely on polygon boundaries. Bubble maps would allow the visualization of local clusters and point-specific trends that are not visible in aggregated data.

## Heatmap

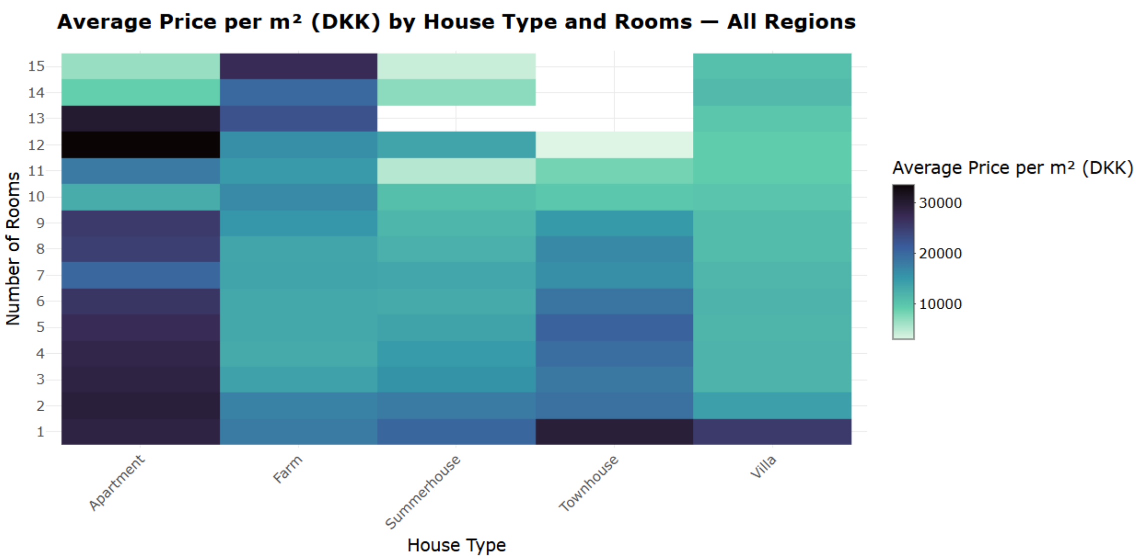
One specific type of visualization that we wanted to include in our project was the heatmap, since they are particularly good at revealing patterns amazingly fast to the viewer. This makes for a useful way to convey patterns without having to show many numbers that are extremely hard to explain.

In our project, we created a heatmap to show the different effects that the number of rooms has on the market for each house type. We have allowed the user of our visualization to choose between 3 different metrics:

1. Average price per square meter
2. Average purchase price
3. Average year built

These metrics help us to see the relationship between price and number of rooms for each category because they seem to have an impact on house pricing.

This visualization also includes a filter for the different regions of Denmark and a purchase year slider to allow for the exploration of the statistics from specific regions the user wants to observe, as well as different years the houses have been sold.



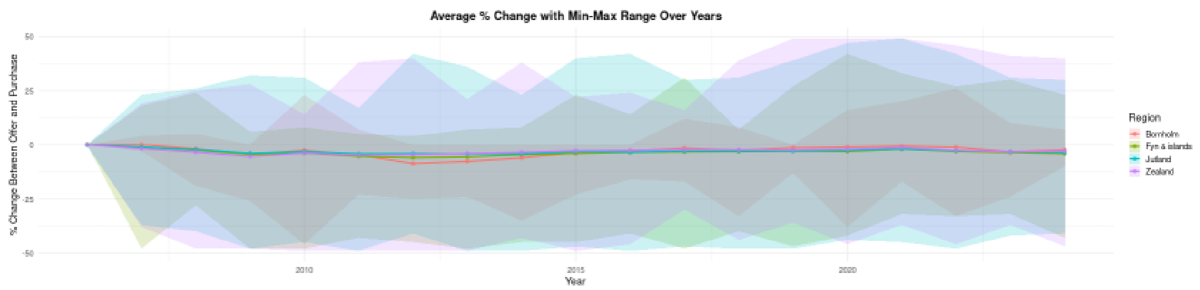
## Price Negotiations

In our dataset, one aspect that was showing and we wanted to analyse is the negotiation price. This is shown in our dataset with the parameter “change_between_offer_and_perchage” which shows the percentage difference between the listed price and the closing price.

In this line plot, we show the mean percentage change throughout the years. It is important to mention that the dataset does not have any official price changes or negotiations before 2006. The average change aside, we want to show the maximum and minimum percentage of price changes that occurred in every year.

The filters that have been included in this graph are a drop-down menu where the user can choose from all available large areas in Denmark and examine only them, as indicated by the different colouring in the legend. The user also has the option to choose which year span they want to observe by using the slider on the left. The final slider is for the user to control the range of square meters that they want to filter the results by.

By comparing the different regions of Denmark, we can observe that the Bornholm region of Denmark has overall the lowest changes in purchasing prices. On the other hand, the highest negotiation changes are in the Jutland and Sjaelland regions of Denmark. This is attributed to the largest percentage of the overall population living in these regions. Finally, by observing the overall available data, we figured that the average negotiations tend to always be in favour of the buyer, with the negotiating price settling on average below the original price. However, we can see that the negotiating price fluctuates a lot, from –50% of the original price to +50% of the original price.




## AI: Price Density Trends

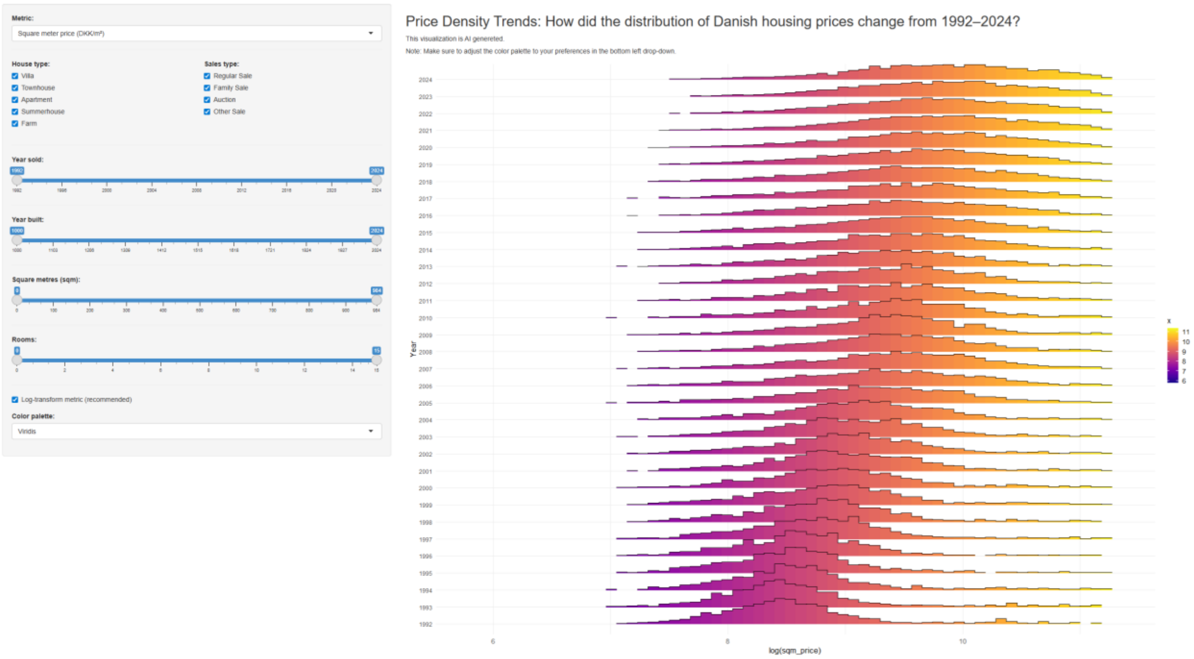
To explore additional perspectives on the housing data, we developed a separate visualization using OpenAI’s GPT-5.1 model. The goal was to investigate whether an AI model could propose a visualization in R Shiny that differed from our own design choices and potentially reveal new analytical insights.

We asked the LLM to “Create a unique visualization of the Danish Residential Housing Prices 1992–2024 data as R code“. To guide the model, we provided contextual information, including a detailed list of all dataset columns with descriptions, a small sample of representative rows, and a minimal Shiny module template. The intention was to enable the model to generate code that could be integrated into our application with minimal modification.

We iterated through several AI-generated proposals, refining our prompts along the way. Early iterations produced a map, but these were either too similar to existing visualizations in our app or did not provide meaningful new insights. After additional prompt refinement, the model proposed a ridgeline density plot, which we ultimately implemented.

## Ridgeline Plot

The ridgeline plot answers the question: How has the distribution of Danish housing prices changed from 1992 to 2024?




The ridgeline plot visualizes the distribution of a selected housing metric for each year. This allows users to observe changes in spread, skewness, and outliers across time.

The AI-generated module includes:

- Metric: Square-meter price, purchase price, house size, number of rooms, and year built.
- Filters: House type, sales type, year sold, year built, size, and number of rooms.
- Log-transformation: Recommended for highly skewed distributions such as prices.
- Color palette selector: Several color palette options.

This visualization demonstrates how AI can assist in the creative and technical stages of visualization design. However, the final implementation required manual refinement and debugging.

# 5. Story/Results

## Purchase Trend

The graphs explain the question: What are the most popular types of houses on sales in Denmark? As the histogram and bar chart show, Villa is the most popular houses on sales, followed by Apartment, Summerhouse, Townhouse, and Farm. It also answered the question: What is the most purchased type of house in Denmark in the past decade? The Box plot and the scatter plot explain how the prices were distributed among different house types. And we could say that, except for Farm, most of the other had a fairly even distribution in terms of Q1 and Q3. The last question: Does the house purchase uprising or downsizing through the years in Denmark? And to explore this, we observe the Box plot with the stacked option selected, and it describes that in 1992 it started with around 1B and steadily increased up to 2021 reaching 20.1B, but after then it decreased sharply to 13.6B and is still decreasing slightly, and at the end of 2024 it became 10.6B.

## Price Index

This tap explains the question: What was the house price index through the years? If we plot the graph on a yearly basis and set 2020 as a benchmark year, then in 1992 it was 36.5%, from there it increased gradually, and in 2007 it became 80.2%, then it flattened slightly until 2013 with 74.2%. After that, the price index increased incrementally, and at the end of 2024, it became 107.3% compared to 2020. It means that the price index is increasing.

## Map

The map explores the question: How do square-meter prices vary by area for Danish residential housing? The visualization answers this clearly by highlighting major urban centres, most notably Copenhagen, as the most expensive area in Denmark. Copenhagen stands out with the highest square-meter prices, followed closely by the other large cities, Aarhus, Odense, and Aalborg, where the city centres are especially prominent on the map.

The map also reflects a well-known Danish stereotype: the wealthy areas north of Copenhagen. These zip codes appear distinctly more expensive than most other regions, confirming their reputation.

A broader pattern is visible as well, nearly any zip code containing a larger town, rather than just a small village, shows elevated prices.

Finally, the map hints at the Danish preference for coastal summerhouses. Several coastal zip codes, despite lacking major cities, still show relatively high square-meter prices, suggesting that proximity to the coast increases housing value even in less urban areas.

## Heatmap

As discussed in the relevant section, the heatmap is good at revealing irregularities and patterns very easily. The research question we want to answer with this visualization is: Does the number of rooms affect the sales of houses in Denmark, and if so, then how? When examining the heatmap with all filters applied, thus considering the entire Danish housing market, we can recognize some patterns for the price per square meter and the purchase price.

Across all housing types, the price for each square meter tends to decrease when the number of rooms increases. There are some cases, like with apartments and farms, where after a certain number of rooms, the price increases again dramatically, but it decreases again for apartments, while for farms it starts to increase after the number of rooms is higher than 12.

If we then examine the average purchase price, then we have a completely different story. The average purchase price tends to increase, which is contrasted with the decrease in average price per square meter. We can assume that the rate at which the average price per square meter does not scale with the rate at which the size of these houses increases, making the houses more expensive overall. Therefore, we can conclude that the number of rooms in a house can impact the purchase price a lot.

## Price Negotiation

For the price negotiation, one thing we need to clarify before we analyze the graph is that we do not have any available data before 2006. From the line graph, it is evident that the average negotiation price does not change a lot. So, on average, the final purchase price stays normal or, at the very least, remarkably close to where the seller wants to stay. However, one interesting observation is that the negotiation price, on average, rarely went above zero. The only case where it is above zero is in the Zealand region, where in 2007 it was +0.1%. This means that the prices are usually negotiated in favor of the buyer rather than the seller. But from the line graph, we can see that there are huge gaps between the minimum and maximum negotiation price and the average. There are cases where a household was sold for almost 1.5 times its original value, as well as cases where a household was sold for almost half of its original value. Overall, this information shows that the market tends to favor the sellers, which includes the average normal family and citizens as well.

## AI: Price Density Trends

The ridgeline plot shows the distribution of Danish residential housing prices for each year from 1992 to 2024. Over time, the entire distribution shifts steadily to the right, indicating an increase in housing prices across the market. This shift is gradual during the 1990s and early 2000s, accelerates in the mid-2000s, and resumes strongly after 2012 following a temporary stabilization around the financial crisis period.

In addition to rising price levels, the distributions become wider in later years, suggesting increased dispersion in housing prices. This indicates growing differences between lower and higher priced segments of the housing market. By the late 2010s and early 2020s, the density peaks at higher price levels than in earlier decades, reflecting both higher typical prices and a larger share of high-priced properties.

The visualization highlights an upward trend in Danish residential housing prices over the past three decades.

# 6. Conclusion/Discussion

We have observed outliers in the year built variable, as a small number of data points list the year built as exactly year 1000. These data points create a large discontinuity compared to the rest of the data, where the majority of houses were built after the year 1500. The most likely explanation is that these extreme values function as placeholders or default entries when the actual construction year is missing or unknown. Their presence can distort or skew visualizations. Therefore, identifying and appropriately handling these outliers, either by filtering them out or treating them as missing data, is important to ensure the validity of the visualizations.

It is important to note that this project is based on a 100,000-row sample of the full dataset. While the sample is large enough to be representative and suitable for this project's visualization purposes, future projects could benefit from using the complete dataset to further validate the observed patterns and enable more detailed analysis.

The visualizations developed throughout this project collectively reveal trends and answer the research questions about the Danish housing market. By combining multiple visualization types, such as histograms, box plots, heatmaps, maps, etc. we were able to examine trends, distributions, and spatial patterns that would be difficult to identify from the raw data alone.

From the visualizations, we see an upward trajectory in housing prices from 1992 to 2024, with periods of accelerated growth, temporary stabilization around the financial crisis, and renewed increases in recent years. Villas consistently emerge as the most purchased housing type, suggesting a stable preference for this category across decades.

The spatial visualizations highlight clear regional differences in the Danish housing market, with higher prices concentrated in major urban areas such as Copenhagen and other large cities, as well as in selected coastal and suburban regions. The heatmap analysis indicates that the number of rooms influences housing prices differently across house types, where price per square meter generally decreases as homes become larger, while total purchase prices tend to increase.

The price negotiation visualizations further show that negotiations are on average in favour of buyers, although the magnitude of negotiation varies significantly across regions and time periods. Finally, the AI ridgeline density visualization demonstrates how price distributions have shifted and widened over time, revealing increasing price dispersion alongside rising overall price levels.

# References

[1] Wikipedia, “List of countries and dependencies by area,” 12 12 2025. [Online]. Available: https://en.wikipedia.org/wiki/List_of_countries_and_dependencies_by_area. [Accessed 18 12 2025].

[2] M. Frederiksen, “Danish Residential Housing Prices 1992-2024,” Kaggle, [Online]. Available: https://www.kaggle.com/datasets/martinfrederiksen/danish-residential-housing-prices-1992-2024/data. [Accessed 11 12 2025].

[3] Shiny, “Easy web apps for data science without the compromises,” [Online]. Available: https://shiny.posit.co/. [Accessed 18 12 2025].

[4] Neogeografen, “dagi,” GitHub, [Online]. Available: https://github.com/Neogeografen/dagi. [Accessed 11 12 2025].